



Moisture induced stresses in cross laminated timber (CLT) made from hydrothermally modified wood

Behbood Mohebbi¹ · Vahid Broushakian¹

Received: 29 June 2021 / Accepted: 9 May 2022 / Published online: 11 June 2022

© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2022

Abstract

In the current research work, moisture gradients and moisture induced stresses (MiS) were investigated in cross laminated timber (CLT) bearing hydrothermally treated fir wood in its outermost laminations. Wooden blocks were cut from fir wood planks and hydrothermally treated in a stainless-steel reactor at 170 °C for a holding time of 30 min. The panels were made with combinations of untreated and treated wood with one as well as two treated layers laid up on both sides of the CLT samples. Afterwards, the CLT panels were exposed to drying (from RH 88 to 32%) and wetting (from RH 32 to 88%) cycles to provide moisture gradients as well as moisture induced stresses. The results showed that higher moisture gradients and moisture induced stresses were recorded on the untreated panels. It was revealed that the moisture gradient due to wetting and drying cycles was reduced because of the hydrothermal treatment of wood.

1 Introduction

Cross laminated timbers (CLT panel) are innovative engineered wood products which are made from wood laminations arranged at a 90° angle to each other (Gereke et al. 2010). This product is usually made from massive wood. Therefore, one of the most important disadvantages of these panels is the hygroscopic behavior of wood. When the cross laminated timber (CLT) is exposed to wet or dry climate, similar to wood it can also absorb or desorb moisture from the surrounding climate leading to apparent stresses. These stresses are so-called moisture induced stresses (MISs) that cause cracks in the panels (Angst and Malo 2012; Fragiaco et al. 2011; Gereke et al. 2009; Jönsson 2005; Jönsson and Thelandersson 2003). The stresses can be categorized as compressive or tensile stresses which are commonly referred to as (MISs). The tensile stresses are especially critical for wood based products. It is well known that the tensile strength of timber perpendicular to the grain is very low, and therefore, if the latter is exceeded, cracks may occur on the panels (Angst and Malo 2012). The cracks can evolve in the

longitudinal direction and reduce the load-bearing capacity of individual timber members significantly.

The moisture induced stresses are problematic for massive wood products. There are different reports indicating that the (MISs) occurred in glulam and caused defects and failures in the products when exposed to loads.

According to the reports, moisture due to related ambient humidity could penetrate into wood and pass through the cross section of a glulam producing moisture induced stresses very slowly over time due to the moisture gradient (Jönsson and Sevénsson 2004; Jönsson 2005; Angst and Malo 2010, 2012, 2013). Jönsson and Thelandersson (2003) studied the moisture gradients and subsequent tensile stresses in the glulam and showed that these stresses exceeded the tensile strength perpendicular to the grain during the wetting step due to reduction in the strength by the moisture absorption. Fragiaco et al. (2011) investigated any effects of ambient relative humidity, size of cross section as well as coatings of the beams and showed that (a) the MISs are influenced by distribution of the moisture throughout the beam cross section and types of related climate; (b) a wide range of the fluctuation in the ambient humidity provides severe MISs in the beams; (c) when the MISs exceed the tensile strength perpendicular to the grain, cracks appear on the cross section of the beams; and finally, (d) application of the coatings reduces these MISs significantly. Angst and Malo (2012) studied the moisture gradients and also moisture induced stresses as well as restrained and released

✉ Behbood Mohebbi
mohebbi@modares.ac.ir

Vahid Broushakian
broushakianv@gmail.com

¹ Department of Wood and Paper Sciences, Tarbiat Modares University, P.O. Box: 356-46414, Noor, Mazandaran, Iran

strains in glulam made from spruce wood due to wetting and drying cycles. They showed that moisture gradients occurred in the beams due to the wetting and drying cycles and similarly the gradients in strains and stresses occurred in the beams. Subsequently, cracks appear on the glulam when the stresses overcome the maximum tensile strength perpendicular to the grain of the wood.

Gereke (2009) used the benefits of the thermal treatment and made CLTs from heat-treated beech wood. Their research work revealed that the thermal treatment of wood reduces stresses due to the moisture variation in the CLT panels.

Therefore and also according to the reports, one of the techniques that might control moisture behavior in wood is the heat treatment that leads to improved dimensional stability (Borysiuk et al. 2014; Mirzaei et al. 2012; Mohebbi et al. 2008). The idea of application of hydrothermally modified wood was recently examined by Mirzaei et al. (2017, 2018) for glulam. They made glulam beams from hydrothermally treated wood and then exposed them to wetting and drying cycles. For the drying cycles, the specimens were initially exposed to RH 90% and then to RH 50%, and for the wetting cycles, the sample beams were exposed to RH 50% and then RH 90%. The results showed that the moisture induced stresses were successfully reduced in the hydrothermally treated beams. Gereke et al. (2009) also made CLT panels from untreated and treated beech wood and then exposed them to RH 65% and then RH 35%. Their results also showed low moisture induced stresses as well as moisture gradients in the treated CLT panels. In the current research work, the main objective was to find out whether one and two layers of hydrothermally treated wood on both sides of the CLT panels could act as moisture protecting elements to reduce moisture gradients and subsequent moisture induced stresses in the CLT panels or not.

2 Materials and methods

2.1 Wood

Wood blocks were prepared with dimensions of 50 (R) × 150 (T) × 550 (L) mm from fir wood (*Abies* sp.) lumber.

2.2 Hydrothermal treatment

Wood blocks were initially placed in a stainless steel cylinder containing water and treated at 170 °C for a holding time of 30 min. Afterwards, the treated wood blocks were initially air seasoned and then dried in a vacuum dryer at 60 °C for 72 h to achieve a moisture content of 10% prior to manufacturing the CLT panels.

2.3 Manufacture of the CLT panels

Lamina with dimensions of 20 (t) × 50 (w) × 550 (l) mm were cut from the blocks and planned into required sizes and then laid up according to Table 1. A two-component polyurethane adhesive (200 g/m²) was applied on the panels. In the treated panels, one and two plies were arranged at both sides according to Table 1. Finally, the CLT panels were pressed for 24 h in a hydraulic press under a pressure of 10 bar. The sample size of the panels was 100 (t) × 550 (w) × 550 (l) mm and at least 4 panels were prepared for each treatment. The sample panels had a moisture content of 12% prior to the test.

Table 1 Assembly of untreated and treated CLT panels

	Total thickness of CLT (mm)	Thickness of layers (mm)	Untreated layers at ply no.	Treated layers at ply no.	Number of treated layers	Total layers in CLT panel
	100	20	2-3-4	1-5	2	5
	100	20	3	1-2-4-5	4	5
						Ply 1 Ply 2 Ply 3 Ply 4 Ply 5

2.4 Experiments

2.4.1 Moisture gradient

The CLT panels were divided into two groups of wetting and drying. For wetting cycles, the conditioned specimens were initially exposed to a climate chamber with RH 32% and thereafter to RH 88%; for the drying cycles, the specimens were initially exposed to RH 88% and thereafter to RH 32%. Each wetting or drying step lasted up to 90 days. Afterwards, the specimens were taken out of the climate chambers and then the CLT panels were cut from adhesive bond lines to provide test slices with 450 (L) × 45 (W) × 20 (T) mm from the whole overall CLT panels in all 5 layers. Afterwards, the slices were weighed and then dried in an oven at 103 ± 2 °C for 24 h, and finally weighed again to determine the dry weight. Moisture contents of the slices were calculated according to Eq. (1) to draw moisture gradients in the panels.

$$MC = \frac{M_1 - M_2}{M_2} \times 100 \quad (1)$$

where MC moisture content (%), M_1 initial weight prior to drying in the oven (g), M_2 final dry weight after drying in the oven (g).

2.4.2 Moisture induced strain gradient in CLT panel

2.4.2.1 Restrained strain gradient For determination of the restrained strain, the panels were initially divided into two groups and then one group of the CLT panels was initially exposed to a drying cycle (RH 88–32%) and another group to a wetting cycle (RH 32–88%) for 45 days. Two points were initially marked at both ends of the odd layers (plies 1, 3, 5) of the panels, and the length between both points was measured and recorded as L_0 , and then the CLT panels were exposed to the next step of wetting (RH 88%) or drying (RH 32%). At the end of each step, the dimensions were measured again and the test panels were exposed to the further reverse steps of drying (RH 32%) or wetting (RH 88%) cycles. For example, in the drying cycle, the measured dimensions were initially determined and recorded as L_0 , and then they were exposed to RH 88% as wetting step.

At the end of this step, the dimensions were measured and recorded as L_1 . Finally the restrained strains were calculated according to Eq. (2) (Gereke et al. 2009).

2.4.3 Released strain gradient

After wetting or drying cycles and measuring the released strain, the specimens were taken out of the climate chamber and then cut from adhesive bond lines to measure the released strain by recording the length between the marked points as L_2 . The released strain was also calculated using Eq. (3) according to Gereke et al. (2009).

$$\varepsilon_{RS} = \frac{L_1 - L_0}{L_0} \quad (2)$$

$$\varepsilon_r = \frac{L_2 - L_1}{L_1} \quad (3)$$

where ε_{RS} restrained strain (mm/mm), L_0 length between two points before exposure to wetting or drying cycles (mm), L_1 length between two points after exposure to reverse cycles (mm); ε_r released strain (mm/mm), L_2 length between two points after cutting the CLT panels from bond lines.

2.4.4 Bending static test gradient

The test panels were cut into specimens (450 (L) × 45 (W) × 20 (T) mm) adopted from ASTM D 143, and then the 3-point bending test was performed using a Dartec testing machine (Fig. 1). Displacement increments were applied at a rate of 4 mm/min. Modulus of elasticity (MOE) and rupture (MOR) of the CLT panel layers were calculated according to Eqs. (4) and (5).

$$MOE = \frac{L^3 (P_{40\%} - P_{10\%})}{4WB^3 (X_{40\%} - X_{10\%})} \quad (4)$$

$$MOR = \frac{3LP_{\max}}{2WB^2} \quad (5)$$

where MOE modulus of elasticity (N/mm²), MOR modulus of rupture (N/mm²), $P_{40\%}$ and $P_{10\%}$ loads at 40% and 10% of max load (N), W and B the width and specimens (mm), L

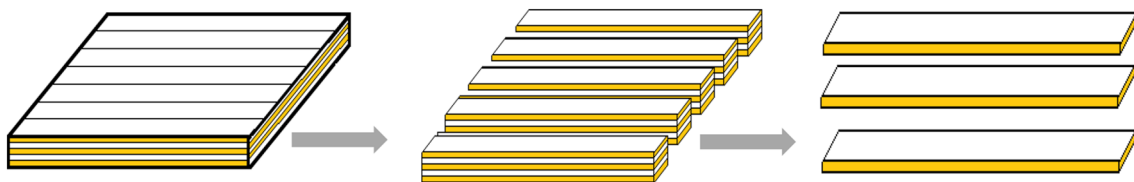


Fig. 1 Preparation of samples for bending test

span of the laminations (mm), $X_{40\%-10\%}$ deflection at 40 and 10% max load (mm).

2.4.5 Moisture induced stress gradient

The moisture induced stresses were calculated according to the data obtained for the MOE and the restrained strain by using Eq. (6) (Angst and Malo 2012).

$$\sigma = \varepsilon_r \times E \quad (6)$$

where σ moisture induced stresses (N/mm²), ε_r restrained strain (mm/mm), E is the modulus of elasticity of each odd layer (N/mm²).

3 Results and discussion

3.1 Moisture gradient

Moisture gradient curves of the CLT panels are shown in Fig. 2. According to the results, any gradient of the moisture is clearly reduced in the panels with 1 or 2 plies of hydrothermally treated wood; in particular, in those layers where the hydrothermally treated wood was used. However, those panels with totally untreated plies showed gradients with higher moisture content indicating that there are differences in moisture contents between mid and outermost layers in the CLT panels. In the treated panel, those plies made with untreated wood showed higher moisture contents in contrast

to the treated plies. Any increase or decrease in the moisture content causes severe dimensional changes due to swelling or shrinkage. Therefore, it is likely that application of the treated plies to the CLT could reduce moisture changes and subsequent dimensional stability.

It can be stated that any thermal treatment increases hydrophobicity of wood and consequently the dimensional stability. Earlier research reports dealing with the thermal treatment of wood confirm the mentioned phenomenon (Kesik et al. 2014; Korkut et al. 2013; Mohebbi and San-aei 2005; Gereke et al. 2009, 2010). The hydrophobicity of treated wood is due to hemicellulose degradation or removal from the cell wall, reduction in hydroxyl groups of cell wall polymers, and increased crystallinity of cellulose (Tjeerdsma and Militz 2005; Poncsák et al. 2006; Yildiz and Gümüşkaya 2007; Kök 2010; Weiland and Guyonnet 2003; Cao et al. 2012; Olarescu et al. 2014; Gündüz et al. 2009).

3.2 Restrained- and released-strain

Moisture induced strain gradients as the restrained- and released-strains in CLT panels are shown in Fig. 3. In wet to dry cycle, the restrained gradient was reduced in the panels and plies with hydrothermally treated wood (Fig. 3a). However, in the treated panels, the mid layer showed higher restrained gradients. After cutting the panels, it was revealed that the panels with the treated plies had very low released strain (Fig. 3a). The low released strain indicates less stresses due to moisture gradients might have occurred in the plies. For the dry to wet cycle,

Fig. 2 Moisture gradients after wet to dry (RH 88–32%) (a, b) and dry to wet cycles (RH 32–88%) (c, d) in untreated, treated one- and two-ply CLT panels; 1-ply indicates layers 1 and 5 are treated and 2-ply indicates layers 1, 2, 4 and 5 are heat treated

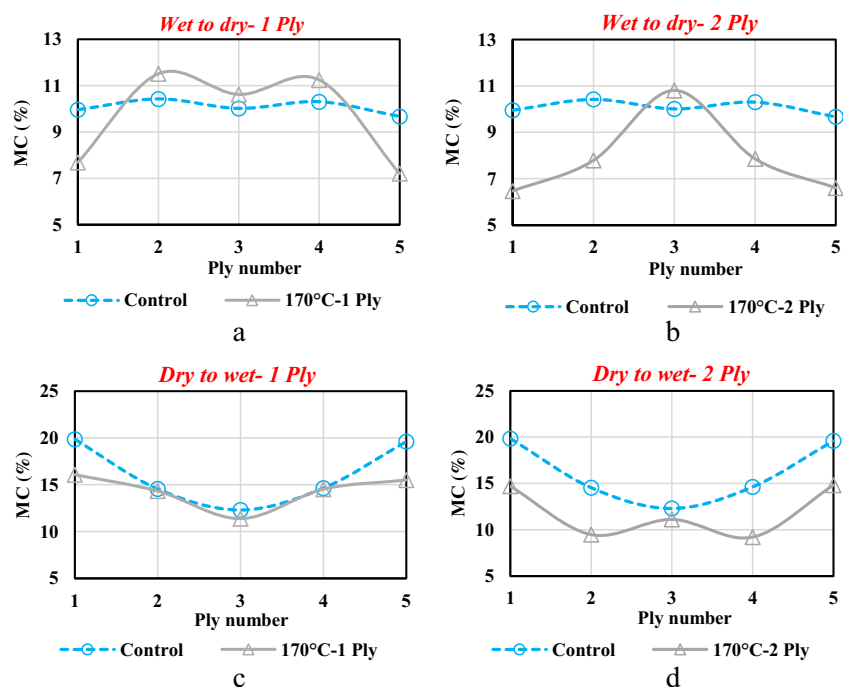
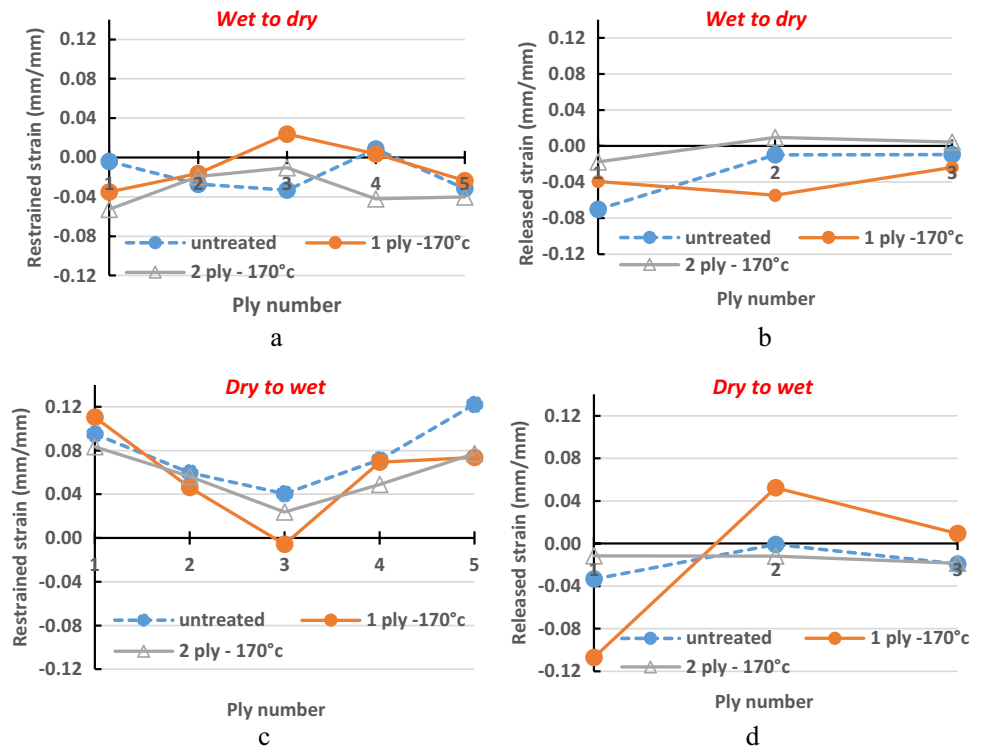


Fig. 3 Restrained strain and released strain after wet to dry cycle (RH 88–32%) (a, b) and dry to wet cycle (RH 32–88%) (c, d), in untreated; treated one- and two-ply CLT panels; 1-ply indicates layers 1 and 5 are treated and 2-ply indicates layers 1, 2, 4 and 5 are heat treated



the gradients for all panels were clear (Fig. 3c). However, the amount of restrained strains in the untreated panel was significantly higher than that of the treated ones. In plies with treated wood, the restrained strain was less. It was revealed that the mid layer of the treated panels with not hydrothermally treated wood showed lesser strains. It seems that the outermost layers played a protective role for the mid layer to be affected by moisture absorption (Fig. 3c). Except for the treated panels with one ply of treated wood on both sides, other treated panels showed no noticeable gradients in the panels. However, in the untreated one, there was a released strain gradient. The lower released strain is due to the lower moisture induced stresses in plies with hydrothermally modified wood.

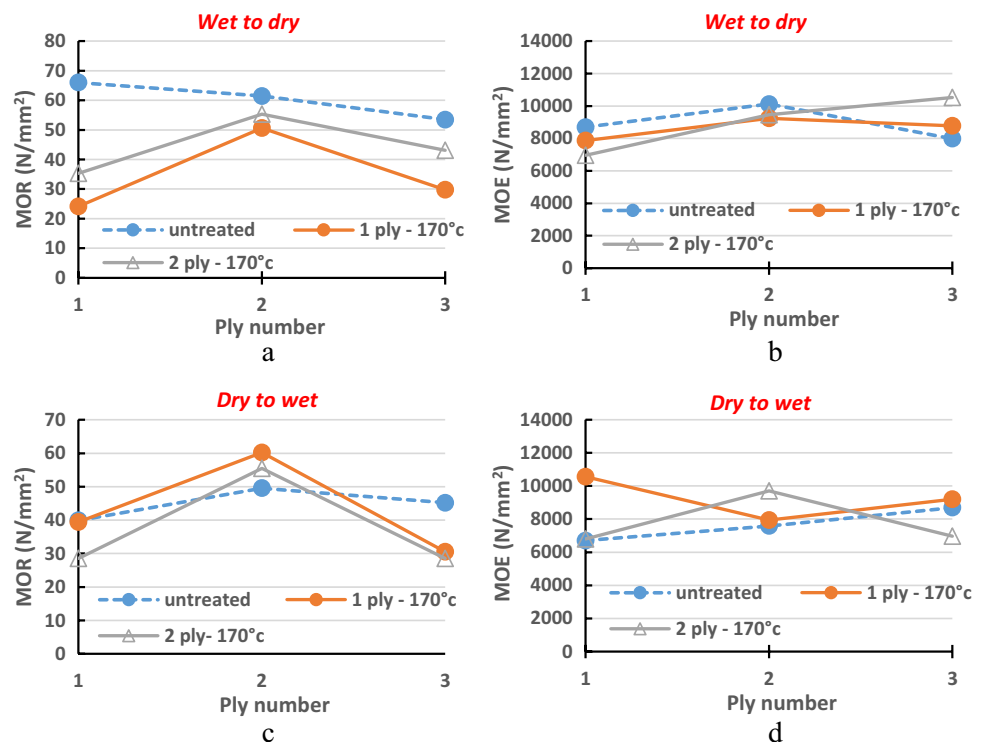
As known, the physical and mechanical properties of wood are governed by the constituent polymers in wood cell walls, cellulose, hemicellulose and lignin. These chemical components are affected by treatment temperature and heat treatment time. As explained above, hemicelluloses which are mostly hydrophilic polymers in wood are more susceptible to thermal degradation. Therefore, any removal of those polymers as well as reduction in the hydroxyl groups in other polymer increases the hydrophobicity of wood. Any reduction in the hydrophilic properties of wood could affect the moisture absorption and desorption and consequently the moisture induced stresses in wood (Bekhta and Niemz 2003; Esteves et al. 2008; Korkut and Guller 2008; Esteves and Pereira. 2008; Aksoy et al. 2011; Percin et al. 2016).

3.3 Modulus of elasticity and rupture in CLT panel layer

Modulus of rupture (MOR) as well as the modulus of elasticity (MOE) for CLT panels exposed to wet to dry (RH 88–32%) and dry to wet (RH 32–88%) cycles are presented in Fig. 4. The MOR was slightly reduced in the plies with treated laminations and small gradients occurred in those CLT panels after wet to dry cycles (Fig. 4a). However, there were no differences between the CLT panels with treated and untreated plies (Fig. 4b). Similarly, the MOR was slightly reduced in the CLT panels with the treated plies (Fig. 4c), and no differences were determined in the CLT panels after dry to wet cycles (Fig. 4d). The slight reduction in the MOR and MOE could be related to strength loss effects in wood due to the hydrothermal treatment. Most of the reports indicated the strength properties of heat-treated wood are reduced after any treatment (Percin et al. 2016).

Bengtsson et al. (2002) reported that the modules of elasticity is decreased by almost 5.3% after heat treatment. Esteves and Pereira (2008) showed that with a mild and lower temperature quite higher mechanical properties can be achieved in thermally modified wood. Anyhow, in general, thermal degradation as well as further reactions in the cell wall polymers are the reasons for any reductions in moduli of elasticity and rupture (Percin et al. 2016; Esteves et al. 2008; Kotilainen et al. 2000). Among the cell wall polymers, cellulose, hemicelluloses and lignin, the hemicelluloses have

Fig. 4 Bending modulus of elasticity (MOE) and modulus of rupture (MOR) after wet to dry (RH 88–32%) (a, b) and dry to wet cycles (RH 32–88%) (c, d) in untreated; treated one- and two-ply CLT panels; 1-ply indicates layers 1 and 5 are treated and 2-ply indicates layers 1, 2, 4 and 5 are heat treated



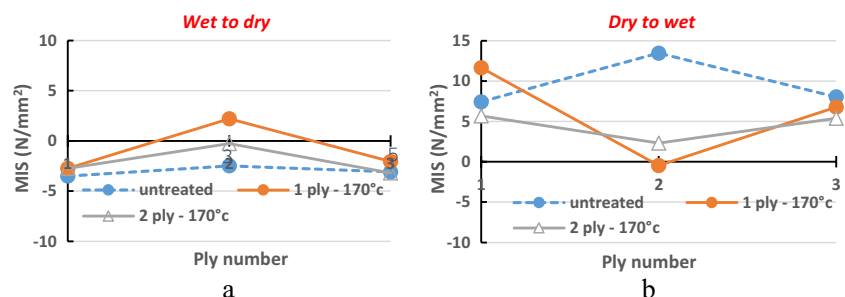
less resistance to heat and they are degraded at higher temperatures during the thermal wood modification. Any loss in hemicelluloses can affect the mechanical properties of wood (Biziks et al. 2013).

3.4 Moisture induced stresses in CLT panels

Moisture induced stresses of the CLT panels after wet to dry (RH 88–32%) and dry to wet (RH 32–88%) cycles are presented in Fig. 5. Results revealed that there were no noticeable gradients nor differences between the treated and untreated panels after wet to dry cycles (Fig. 5a). However, it was revealed that there were increased moisture induced stresses in the CLT panels with untreated laminations as well as panels with only one ply of hydrothermally treated wood at both surface layers and the gradients that were also determined in both types of the CLT panels (Fig. 5b). It was interesting to note that the

stresses were lower in the mid layer of the CLT panels with one-ply of treated wood; however, it was the highest in the mid layer of the untreated panels after dry to wet cycle (Fig. 5b). In addition, panels with two-plyes of hydrothermally treated wood showed the lowest moisture induced stresses. It can be stated that the application of heat-treated wood to outermost layers could play a hydrophobic protective role for the mid layers of the CLT panels and they were also exposed to lesser swelling during wetting conditions. After absorbing any moisture during dry to wet cycle, the panels could resist to shrinkage due to dimensional stability of the treated laminations in the CLT panels. Most of the reports indicated less moisture absorption as well as higher dimensional stability of the heat-treated wood (Gereke et al. 2009, 2010; Percin et al. 2016; Esteves et al. 2008; Kotilainen et al. 2000; Mohebbi and Sanaei 2005; Mohebbi et al. 2008; Mirzaei et al. 2012, 2017).

Fig. 5 Moisture induced stresses (MIS) in CLT panels after wet to dry (RH 88–32%) (a) and dry to wet cycles (RH 32–88%) (b) in untreated; treated one- and two-ply CLT panels; 1-ply indicates layers 1 and 5 are treated and 2-ply indicates layers 1, 2, 4 and 5 are heat treated



4 Conclusion

In the current research work the idea was to apply heat-treated wood on the outermost layers of CLT panels to overcome and reduce moisture induced stresses. In the panels with five plies, one or two plies of the hydrothermally treated wood were used at both sides and compared with the untreated ones. According to the results, the following conclusions can be drawn:

- Application of heat-treated wood on the outermost layer of CLT reduces the moisture gradient in the panels. The panels with two plies at both sides show better reduction in moisture gradients than one-ply panels.
- In the CLT panels with hydrothermally treated layers, the restrained and released stresses are reduced. It is more prominent in the panels with the two plies than one ply.
- There are no big differences between moduli of elasticity and rupture of the panels with treated and untreated laminations.
- Any application of heat-treated wood in CLTs reduces moisture induced stresses.

Author contributions VB took primary responsibility for planning the work, collecting the literature, and writing the original draft of the manuscript. BM instructed and supervised the research process as well as reviewed and edited the final version of the manuscript.

Declarations

Conflict of interest The authors declare no conflicts of interests. The founding sponsor had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript, and in the decision to publish the results.

References

- Aksoy A, Deveci M, Baysal E, Toker H (2011) Color and gloss changes of Scots pine after heat modification. *Wood Res* 56(3):329–336
- Angst V, Malo KA (2010) Moisture induced stresses perpendicular to the grain in glulam: review and evaluation of the relative importance of models and parameters. *Holzforschung* 64(5):609–617
- Angst V, Malo KA (2012) The effect of climate variations on glulam-an experimental study. *Eur J Wood Prod* 70(5):603–613
- Angst V, Malo KA (2013) Moisture-induced stresses in glulam cross sections during wetting exposures. *Wood Sci Technol* 47(2):227–241
- Bekhta P, Niemz P (2003) Effect of high temperature on the change in color, dimensional stability and mechanical properties of spruce wood. *Holzforschung* 57(5):539–546
- Bengtsson C, Jermer J, Brem F (2002) Bending strength of heat-treated spruce and pine timber. In: Annual meeting international research group on wood preservation, Proceedings, Stockholm, vol 33
- Biziks V, Andersons B, Beļkova L, Kapača E, Militz H (2013) Changes in the microstructure of birch wood after hydrothermal treatment. *Wood Sci Technol* 47(4):717–735
- Borysiuk P, Boruszewski P, Radosław A, Marcin G (2014) Dimensional stability of layered elements made of thermally modified wood. *Trieskové Beztrieskové Obrábanie Dрева* 9(1):191–196
- Cao Y, Lu J, Huang R, Jiang J (2012) Increased dimensional stability of Chinese fir through steam-heat treatment. *Eur J Wood Prod* 70(4):441–444
- Esteves B, Pereira H (2008) Wood modification by heat treatment: a review. *BioResources* 4(1):370–404
- Esteves B, Domingos I, Pereira H (2008) Pine wood modification by heat treatment in air. *BioResources* 3(1):142–154
- Fragiacomo M, Fortino S, Tononi D, Usardi I, Toratti T (2011) Moisture-induced stresses perpendicular to grain in cross-sections of timber members exposed to different climates. *Eng Struct* 33(11):3071–3078
- Gereke T (2009) Moisture-induced stresses in cross-laminated wood panels, Ph.D. Thesis, ETH, Zurich, p 193
- Gereke T, Schnider T, Hurst A, Niemz P (2009) Identification of moisture-induced stresses in cross-laminated wood panels from beech wood (*Fagus sylvatica* L.). *Wood Sci Technol* 43(3):301–315
- Gereke T, Hass P, Niemz P (2010) Moisture-induced stresses and distortions in spruce cross-laminates and composite laminates. *Holzforschung* 64(1):127–133
- Gündüz G, Aydemir D, Karakas G (2009) The effects of thermal treatment on the mechanical properties of wild Pear (*Pyrus elaeagnifolia* Pall) wood and changes in physical properties. *Mater Des* 30(10):4391–4395
- Jönsson JH (2005) Internal stresses in glulam due to moisture gradients in the grain direction. *Holzforschung* 59(1):18–22
- Jönsson JH, Sevénsson S (2004) Internal stresses in the cross-grain direction in glulam induced by climate variations. *Holzforschung* 58(2):154–159
- Jönsson J, Thelandersson S (2003) The effect of moisture gradients on tensile strength perpendicular to grain in glulam. *Holz Roh-Werkst* 61(5):342–348
- Kesik HI, Korkut S, Hiziroglu S, Sevik H (2014) An evaluation of properties of four heat treated wood species. *Ind Crop Prod* 60:60–65
- Köl HS (2010) Characteristics of heat-treated Turkish pine and fir wood after ThermoWood processing. *J Environ Biol* 31(6):1007–1011
- Korkut DS, Guller B (2008) The effects of heat treatment on physical properties and surface roughness of red-bud maple (*Acer trautvetteri* Medw) wood. *Bioresource Technol* 99(8):2846–2851
- Korkut DS, Hiziroglu S, Aytin A (2013) Effect of heat treatment on surface characteristics of wild cherry wood. *BioResources* 8(2):1582–1590
- Kotilainen RA, Toivanen TJ, Alén RJ (2000) FTIR monitoring of chemical changes in softwood during heating. *J Wood Chem Technol* 20(3):307–320
- Mirzaei G, Mohebby B, Tasooji M (2012) The effect of hydrothermal treatment on bond shear strength of beech wood. *Eur J Wood Prod* 70(5):705–709
- Mirzaei G, Mohebby B, Ebrahimi G (2017) Glulam beam made from hydrothermally treated poplar wood with reduced moisture induced stresses. *Constr Build Mater* 135:386–393
- Mirzaei G, Mohebby B, Ebrahimi G (2018) Technological properties of glulam beams made from hydrothermally treated poplar wood. *Wood Mater Sci Eng* 13(1):36–44
- Mohebby B, Sanaei I (2005) Influences of the hydro-thermal treatment on physical properties of beech wood (*Fagus orientalis*). In: 36th annual meeting, Bangalore, India, pp 24–28

- Mohebbi B, Ilbeighi F, Kazemi-Najafi S (2008) Influence of hydrothermal modification of fibers on some physical and mechanical properties of medium density fiberboard (MDF). *Holz Roh-Werkst* 66(3):213–218
- Olarescu MC, Campean M, Ispas M, Cosoreanu C (2014) Effect of thermal treatment on some properties of lime wood. *Eur J Wood Prod* 72(4):559–562
- Percin O, Peker H, Atilgan A (2016) The effect of heat treatment on the some physical and mechanical properties of beech (*Fagus orientalis* lipsky) wood. *Wood Res* 61(3):443–456
- Poncsák S, Kocaefe D, Bouazara M, Pichette A (2006) Effect of high temperature treatment on the mechanical properties of birch (*Betula papyrifera*). *Wood Sci Technol* 40(8):647–663
- Tjeerdsma BF, Militz H (2005) Chemical changes in hydrothermal treated wood: FTIR analysis of combined hydrothermal and dry heat-treated wood. *Holz Roh-Werkst* 63(2):102–111
- Weiland JJ, Guyonnet R (2003) Study of chemical modifications and fungi degradation of thermally modified wood using DRIFT spectroscopy. *Holz Roh-Werkst* 61(3):216–220
- Yildiz S, Gümüşkaya E (2007) The effects of thermal modification on crystalline structure of cellulose in soft and hard wood. *Build Environ* 42(1):62–67

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Terms and Conditions

Springer Nature journal content, brought to you courtesy of Springer Nature Customer Service Center GmbH (“Springer Nature”).

Springer Nature supports a reasonable amount of sharing of research papers by authors, subscribers and authorised users (“Users”), for small-scale personal, non-commercial use provided that all copyright, trade and service marks and other proprietary notices are maintained. By accessing, sharing, receiving or otherwise using the Springer Nature journal content you agree to these terms of use (“Terms”). For these purposes, Springer Nature considers academic use (by researchers and students) to be non-commercial.

These Terms are supplementary and will apply in addition to any applicable website terms and conditions, a relevant site licence or a personal subscription. These Terms will prevail over any conflict or ambiguity with regards to the relevant terms, a site licence or a personal subscription (to the extent of the conflict or ambiguity only). For Creative Commons-licensed articles, the terms of the Creative Commons license used will apply.

We collect and use personal data to provide access to the Springer Nature journal content. We may also use these personal data internally within ResearchGate and Springer Nature and as agreed share it, in an anonymised way, for purposes of tracking, analysis and reporting. We will not otherwise disclose your personal data outside the ResearchGate or the Springer Nature group of companies unless we have your permission as detailed in the Privacy Policy.

While Users may use the Springer Nature journal content for small scale, personal non-commercial use, it is important to note that Users may not:

1. use such content for the purpose of providing other users with access on a regular or large scale basis or as a means to circumvent access control;
2. use such content where to do so would be considered a criminal or statutory offence in any jurisdiction, or gives rise to civil liability, or is otherwise unlawful;
3. falsely or misleadingly imply or suggest endorsement, approval, sponsorship, or association unless explicitly agreed to by Springer Nature in writing;
4. use bots or other automated methods to access the content or redirect messages
5. override any security feature or exclusionary protocol; or
6. share the content in order to create substitute for Springer Nature products or services or a systematic database of Springer Nature journal content.

In line with the restriction against commercial use, Springer Nature does not permit the creation of a product or service that creates revenue, royalties, rent or income from our content or its inclusion as part of a paid for service or for other commercial gain. Springer Nature journal content cannot be used for inter-library loans and librarians may not upload Springer Nature journal content on a large scale into their, or any other, institutional repository.

These terms of use are reviewed regularly and may be amended at any time. Springer Nature is not obligated to publish any information or content on this website and may remove it or features or functionality at our sole discretion, at any time with or without notice. Springer Nature may revoke this licence to you at any time and remove access to any copies of the Springer Nature journal content which have been saved.

To the fullest extent permitted by law, Springer Nature makes no warranties, representations or guarantees to Users, either express or implied with respect to the Springer nature journal content and all parties disclaim and waive any implied warranties or warranties imposed by law, including merchantability or fitness for any particular purpose.

Please note that these rights do not automatically extend to content, data or other material published by Springer Nature that may be licensed from third parties.

If you would like to use or distribute our Springer Nature journal content to a wider audience or on a regular basis or in any other manner not expressly permitted by these Terms, please contact Springer Nature at

onlineservice@springernature.com